

INFOE 3: LAB 3_Suppression

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BS INFO 3C

The screenshot displays the Visual Studio Code (VS Code) interface. The Explorer panel on the left shows a project named 'LAB3_FRANCISCO_MANULAT_L...' with a subfolder 'Lab3_Suppression' containing files 'cleaned_employee_data.csv', 'lab3.py', and 'employee_data.csv'. The main editor window is open to 'employee_data.csv', showing 27 lines of data. The first line is a header: 'name,age,zip_code,salary,medical_risk'. The subsequent 26 lines contain individual records, such as 'Gregory Johnson,66,53336,103200,High' and 'Mr. Jeremy Reed,36,10376,87210,Medium'. The status bar at the bottom indicates 'Ln 1, Col 1', 'Spaces: 4', 'UTF-8', 'CRLF', and 'Plain Text'. The Windows taskbar at the very bottom shows the date and time as '12:40 PM 5/13/2026'.

```
1 name,age,zip_code,salary,medical_risk
2 Gregory Johnson,66,53336,103200,High
3 Darrell Davis,84,09744,133527,High
4 Mrs. Cheryl Saunders,57,19325,63220,Medium
5 Michael Gomez,76,27469,131015,Medium
6 Leslie Stevenson,39,90093,144373,High
7 Kathryn Ellis,74,36979,136921,Low
8 Susan Herman,22,41161,86657,Medium
9 Ashley Brown,77,63719,63856,High
10 Dustin Mueller,39,46933,100380,Medium
11 April Zimmerman,55,47228,86121,Low
12 Holly Brown,41,70651,75090,Low
13 Jasmine Shaw,78,80088,130058,High
14 Dr. Kristi Vazquez,61,92274,88611,Medium
15 Courtney Malone,29,57662,61727,High
16 Holly Martin,66,81500,85943,High
17 Mrs. Crystal Blackwell DVM,28,75835,114623,High
18 Carol Tapia,57,26223,42245,High
19 Ashley Jacobs,82,74459,39661,High
20 Ann Frederick,70,06557,80020,High
21 Jennifer Frey,71,78666,38733,Medium
22 Emily Lee,73,93655,118726,High
23 Christopher Garcia,76,17234,97427,High
24 Hannah Blair,59,56347,81220,High
25 Mitchell Wolfe,49,58601,121423,High
26 Mr. Jeremy Reed,36,10376,87210,Medium
27
```

Figure 1: Initial Dataset Inspection.

- This screenshot shows the raw `employee_data.csv` file within the VS Code environment. It contains the original 25 records with all five columns: `name`, `age`, `zip_code`, `salary`, and `medical_risk`. This represents the data state before the suppression algorithm is applied to remove Direct Identifiers and High-Risk attributes.

The screenshot displays the Visual Studio Code interface. The Explorer pane on the left shows a project named 'LAB3_FRANCISCO_MANULAT_LABANON_CORTEJO_3C_Info3' with a subfolder 'Lab3_Suppression' containing 'cleaned_employee_data.csv', 'employee_data.csv', and 'lab3.py'. The main editor window shows the 'lab3.py' script. The script is divided into four parts: loading data, protecting original data, defining a hit list, and executing suppression. The terminal at the bottom shows the execution of the script, which successfully reads 'employee_data.csv', processes it, and saves the 'cleaned_employee_data.csv' file. The terminal output includes a verification of the first record and a confirmation that the final CSV has been saved.

```
1 import csv
2 import copy
3
4 # --- PART 1: LOAD THE DATA ---
5 # This matches the "Load the Data" task in your manual
6 raw_data = []
7 try:
8     # Fixed the encoding syntax (used '=' instead of '-')
9     with open('employee_data.csv', 'r', encoding='utf-8-sig') as file:
10         reader = csv.DictReader(file)
11         for row in reader:
12             # Convert salary to int as required by previous labs
13             row['salary'] = int(row['salary'])
14             raw_data.append(row)
15 except FileNotFoundError:
16     print("Error: employee_data.csv not found. Please create it first!")
17
18 # --- PART 2: PROTECT ORIGINAL DATA ---
19 # Using deepcopy ensures the 'original evidence' is not destroyed
20 cleaned_data = copy.deepcopy(raw_data)
21
22 # --- PART 3: DEFINE THE HIT LIST ---
23 # These are the "Direct Identifiers" and "High-Risk" attributes
24 hit_list = ["name", "medical_risk"]
25
26 # --- PART 4: EXECUTE SUPPRESSION ---
27 # This nested loop removes the sensitive keys safely
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell - Lab3_Suppression

```
PS D:\INFOE3 TASKS\Lab3_Francisco_Manulat_Labanon_Cortejo_3C_Info3> cd Lab3_Suppression
PS D:\INFOE3 TASKS\Lab3_Francisco_Manulat_Labanon_Cortejo_3C_Info3\Lab3_Suppression> python lab3.py
Verification (First Record):
{'age': '66', 'zip_code': '53336', 'salary': 103200}

Success! Final CSV has been saved!
PS D:\INFOE3 TASKS\Lab3_Francisco_Manulat_Labanon_Cortejo_3C_Info3\Lab3_Suppression>
```

Figure 2: Implementation and Execution of Data Suppression.

- This screenshot displays the final Python script and its successful execution in the terminal. The code utilizes `copy.deepcopy()` to protect the original data and a nested loop with `pop(key, None)` to suppress sensitive attributes. The terminal output verifies the removal of `name` and `medical_risk` fields while confirming the successful export of the anonymized dataset.

The screenshot shows a Visual Studio Code editor window with the following components:

- EXPLORER:** Displays the file structure with folders `LAB3_FRANCISCO_MANULAT_LABANON_CORTEJO_3C_Info3` and `Lab3_Suppression`. Inside `Lab3_Suppression`, the files `cleaned_employee_data.csv`, `employee_data.csv`, and `lab3.py` are listed.
- EDITOR:** The active file is `cleaned_employee_data.csv` in the `Lab3_Suppression` folder. It contains a CSV file with 27 rows of data. The first row is the header: `age,zip_code,salary`. The subsequent rows contain numerical values for these three columns. Row 21 is highlighted.
- TERMINAL:** Shows the execution of a PowerShell script. The commands and output are:

```
PS D:\INFOE3_TASKS\Lab3_Francisco_Manulat_Labanon_Cortejo_3C_Info3> cd Lab3_Suppression
PS D:\INFOE3_TASKS\Lab3_Francisco_Manulat_Labanon_Cortejo_3C_Info3\Lab3_Suppression> python lab3.py
Verification (First Record):
{'age': '66', 'zip_code': '53336', 'salary': 103200}
Success! Final CSV has been saved!
PS D:\INFOE3_TASKS\Lab3_Francisco_Manulat_Labanon_Cortejo_3C_Info3\Lab3_Suppression>
```

Figure 3: Exported Anonymized Dataset.

- This screenshot displays the final `cleaned_employee_data.csv` file. The output confirms that the suppression process was successful, as the file now contains only the three non-sensitive columns: `age`, `zip_code`, and `salary`. All direct identifiers and high-risk attributes have been removed according to the lab requirements.